Time Value of Money

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A dollar on hand today is worth more than a dollar to be received in the future because the dollar on hand today can be invested to earn interest to yield more than a dollar in the future

Question out Lines:

1. What is time value of money?
2. Why money does have time value?
3. What is compounding?
4. What is discounting?
5. What is annuity?
6. What is present value and future value?
7. What is the 'Rule of 72'?
8. Problem Solving

Time Value of Money

The Time Value of Money

A dollar on hand today is worth more than a dollar to be received in the future because the dollar on hand today can be invested to earn interest to yield more than a dollar in the future.

Time value of money is a fundamental financial principle that states **a dollar today is worth more than a dollar in the future.**
This is due to the potential for that dollar to earn interest or grow over time.

Key factors that influence the time value of money:

- **Interest rates:** Higher interest rates generally make future money worth less today.
- **Inflation:** Inflation reduces the purchasing power of money over time.
- **Risk:** Higher risk investments often offer higher potential returns, but also come with greater uncertainty.

The Time Value of Money concepts will be grouped into two areas: Future Value and Present Value.

Future Value describes the process of finding what an investment today will grow to in the future.

Present Value describes the process of determining what a cash flow to be received in the future is worth in today's dollars.

Why does money have a time value?

Money has a time value. A rupee today is more valuable than a rupee a year hence. Why?

There are several reasons:

1. Individuals, in general, prefer current consumption to future consumption.
2. Capital can be employed productively to generate positive returns. An investment of one rupee today would grow to (1+r) a year hence (r is the rate of return earned on the investment).
3. In an inflationary period a rupee today represents a greater real purchasing power than a rupee a year hence.

Present Value

Present Value describes the process of determining what a cash flow to be received in the future is worth in today's dollars.

Therefore, the Present Value of a future cash flow represents the amount of money today which, if invested at a particular interest rate, will grow to the amount of the future cash flow at that time in the future. The process of finding present values is called Discounting and the interest rate used to calculate present values is called the discount rate. For example, the Present Value of $100 to be received one year from now is $90.91 if the discount rate is 10% compounded annually. This can be demonstrated as follows: (Refer to the Future Value page if you are unfamiliar with the calculations.)

One Year

$90.91(1 + 0.10) = $100 or

$90.91 = $100/(1 + 0.10)

Notice that the Future Value Equation is used to describe the relationship between the present value and the future value. Thus, the Present Value of $100 to be received in two years can be shown to be $82.64 if the discount rate is 10%.

Two Years

$82.64(1 + 0.10)2 = $100 or

$82.64 = $100/(1 + 0.10)2

A pattern should be becoming apparent. The following equation can be used to calculate the Present Value of a future cash flow given the discount rate and number of years in the future that the cash flow occurs. (This equation can be obtained algebraically from the Future Value Equation.)

$$PV = \frac{CF_t}{(1+r)^t}$$

Where

PV = Present Value
CFt = Future Cash Flow which occurs t years from now
r = the interest or discount rate
t = the number of years

Present Value Example

Find the Present Value of $100 to be received 3 years from today if the interest rate is 10%.

Solution:

$$PV = \frac{100}{(1+.10)^3} = \$75.13$$

Future Value

The Future Value of a cash flow represents the amount, at some time in the future, that an investment made today will grow to if it is invested to earn a specific interest rate

For example, if you were to deposit $100 today in a bank account to earn an interest rate of 10% compounded annually, this investment will grow to $110 in one year. This can be shown as follows:

Year 1

$100(1 + 0.10) = $110

At the end of two years, the initial investment will have grown to $121. Notice that the investment earned $11 in interest during the second year, whereas, it only earned $10 in interest during the first year. Thus, in the second year, interest was earned not only on the initial investment of $100 but also on the $10 in interest that was paid at the end of the first year. This occurs because the interest rate in the example is a compound interest rate.

Compound Interest

Under compound interest, interest is earned not only on the initial principal but also on the accumulated interest. Interest begins to be earned on the accumulated interest as soon as it is paid, which occurs at the end of each compounding period. This is in contrast to simple interest, under which interest is only earned on the initial principal.

Valuations should generally be based on compound interest because, after the interest has been paid, the full amount, i.e., the initial principal plus interest, could be withdrawn and reinvested elsewhere. Thus, interest on the new investment would be earned on the full amount.

The interest rate in the example is 10% compounded annually. This implies that interest is paid annually. Thus the balance in the account was $110 at the end of the first year. Thus, in the second year the account pays 10% on the initial principal of $100 and the $10 of interest earned in the first year. Thus, the $121 balance in the account after two years can be computed as follows:

Year 2

$110(1+0.10) = $121 or

$100(1+0.10)(1+0.10) = $121 or

$100(1+0.10)2 = $121

If the money was left in the account for one more year, interest would be earned on $121, i.e., the initial principal of $100, the $10 in interest paid at the end of year 1, and the $11 in interest paid at the end of year 2. Thus the balance in the account at the end of year three is $133.10. This can be computed as follows:

Year 3

$121(1+0.10) = $133.10 or

$100(1+0.10) (1+0.10) (1+0.10) = $133.10 or

$100 (1+0.10)3 = $133.10

A pattern should be becoming apparent. The Future Value of an initial investment at a given interest rate compounded annually at any point in the future can be found using the following equation:

$$FV_t = CF_0(1+r)^t$$

Where

FVt = the Future Value at the end of year t,
CF0 = the initial investment,
r = the annually compounded interest rate, and
t = the number of years.

Future Value Example

Find the Future Value at the end of 4 years of $100 invested today at an interest rate 10%.

Solution:

$$FV_4 = 100(1+.10)^4 = \$146.41$$

Cash Flow Streams

Present Value

The Present Value of a Cash Flow Stream is equal to the sum of the Present Values of the individual cash flows. To see this, consider an investment which promises to pay $100 one year from now and $200 two years from now. If an investor were given a choice of this investment or two alternative investments, one promising to pay $100 one year from now and the other promising to pay $200 two years from now, clearly, he would be indifferent between the two choices. (Assuming that the investments were all of equal risk, i.e., the discount rate is the same.) This is because the cash flows that the investor would receive at each point in time in the future are the same under either alternative. Thus, if the discount rate is 10%, the Present Value of the investment can be found as follows:

Present Value of the Investment

PV = $100/(1 + 0.10) + $200/(1 + 0.10)2

PV = $90.91 + $165.29 = $256.20

The following equation can be used to find the Present Value of a Cash Flow Stream.

$$PV = \sum_{t=0}^{n} \frac{CF_t}{(1+r)^t}$$

Where

PV = the Present Value of the Cash Flow Stream,
CFt = the cash flow which occurs at the end of year t,
r = the discount rate,
t = the year, which ranges from zero to n, and
n = the last year in which a cash flow occurs**.**

Present Value Example

Find the Present Value of the following cash flow stream given that the interest rate is 10%.

Year	0	1	2	3	4
Cash Flow		100	200	200	300

Solution:

$$PV = \frac{100}{(1+.10)^1} + \frac{200}{(1+.10)^2} + \frac{200}{(1+.10)^3} + \frac{300}{(1+.10)^4} = \$611.37$$

Future Value

The Future Value of a Cash Flow Stream is equal to the sum of the Future Values of the individual cash flows. For example, consider an investment which promises to pay $100 one year from now and $200 two years from now. Given that the discount rate is 10%, the Future Value at the end of year 2 of the investment can be found as follows:

Future Value of the Investment

FV2 = $100(1 + 0.10) + $200

FV2 = $110.00 + $200.00 = $310.00

As of year 2, the $100 received at the end of year 1 would have earned interest for one year while the $200 received at the end of year 2 would not yet have earned any interest. Thus, the Future Value at the end of year 2, i.e., immediately after the $200 cash flow was received, is $310.00.

The following equation can be used to find the Future Value of a Cash Flow Stream at the end of year t

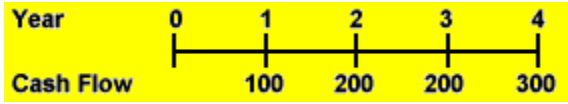
$$FV_n = \sum_{t=0}^{n} CF_t (1+r)^{n-t}$$

Where

FVt = the Future Value of the Cash Flow Stream at the end of year t,
CFt = the cash flow which occurs at the end of year t,
r = the discount rate,
t = the year, which ranges from zero to n, and
n = the last year in which a cash flow occurs.

Future Value Example

Find the Future Value at the end of year 4 of the following cash flow stream given that the interest rate is 10%.



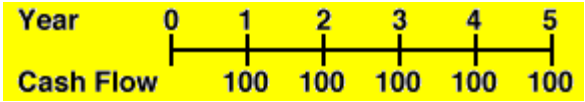
Solution:

$$FV_4 = 100(1+.10)^3 + 200(1+.10)^2 + 200(1+.10)^1 + 300 = \$895.10$$

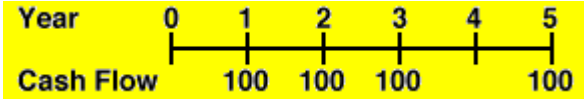
Annuities

An Annuity is a cash flow stream which adheres to a specific pattern.

Namely, an Annuity is a cash flow stream in which the cash flows are level (i.e., all of the cash flows are equal) and the cash flows occur at a regular interval. The annuity cash flows are called annuity payments or simply payments. Thus, the following cash flow stream is an annuity.



While, the following cash flow stream is not an annuity because the payments do not occur at a regular interval.



When a cash flow stream is of the form given in Figure 1, i.e., an annuity, the process of finding the Present Value or Future Value of the cash flow stream is greatly simplified.

Present Value of an Annuity

The Present Value of an Annuity is equal to the sum of the present values of the annuity payments. This can be found in one step through the use of the following equation:

$$PVA = PMT\left[\frac{1-(1+r)^{-t}}{r}\right]$$

Where

PVA = the Present Value of the Annuity
PMT = the Annuity Payment
r = The Interest or Discount Rate
t = The Number of Years (also the Number of Annuity Payments)

Example: Consider the annuity of $100 per year for five years. If the discount rate is equal to 10%, then the Present Value of the Annuity can be found as follows:

Present Value of the Annuity

$$PVA = 100\left[\frac{1-(1+0.10)^{-5}}{0.10}\right] = \$379.08$$

Future Value of an Annuity

The Future Value of an Annuity is calculated at the end of the period in which the last annuity payment occurs.

The Future Value of the Annuity is equal to the sum of the future values of the individual annuity payments at that time.

Thus, the future value of a five year annuity is computed at the end of year five. This can be found in one step through the use of the following equation:

$$FVA_t = PMT\left[\frac{(1+r)^t - 1}{r}\right]$$

Where

FVA = The Present Value of the Annuity
PMT = The Annuity Payment
r = The Interest or Discount Rate
t = The Number of Years (also the Number of Annuity Payments)

Consider the annuity of $100 per year for five years. If the discount rate is equal to 10%, then the Future Value of this Annuity at the end of period five can be found as follows:

Future Value of the Annuity

$$FVA_5 = 100\left[\frac{(1+0.10)^5 - 1}{0.10}\right] = \$610.51$$

What is the 'Rule of 72'?

The 'Rule of 72' is a simplified way to determine how long an investment will take to double, given a fixed annual rate of interest. By dividing 72 by the annual rate of return, investors can get a rough estimate of how many years it will take for the initial investment to duplicate itself.

For example, the rule of 72 states that $1 invested at 10% would take 7.2 years ((72/10) = 7.2) to turn into $2. In reality, a 10% investment will take 7.3 years to double ((1.10^7.3 = 2).

When dealing with low rates of return, the Rule of 72 is fairly accurate. This chart compares the numbers given by the rule of 72 and the actual number of years it takes an investment to double.

Definition of 'Discounting'

The process of determining the present value of a payment or a stream of payments that is to be received in the future. Given the time value of money, a dollar is worth more today than it would be worth tomorrow given its capacity to earn interest. Discounting is the method used to figure out how much these future payments are worth today.

MATH-1:

What is the Present Value of the following Cash Flows at an Interest rate of 12% per year?

I. $100 received today?

II. $100 received after 5 Years from now?

III. $100 received each year beginning 1 year from now and ending 10 years from now?

MATH-2:

What is the value of the following investments: 5 years from now at an Interest rate of 8% per year depending on the Compounding Period?

I. $100 invested today with interest Compounded Annually.

II. $100 invested today with interest Compounded Quarterly.

III. $100 invested today with interest Compounded Monthly.

MATH-3:

Find the Future Value of the following investment at an Interest rate of 8% per year, Compounded Annually.

I. $100 is invested each year beginning 1 year from now and continues through Year 10, when the Proceeds are withdrawn.

II. $100 is invested each year starting today and continuing through Year 10, when the Proceeds are withdrawn.

III. $100 is invested each year beginning 1 year from now and continues through Year 9, when the Proceeds are to be withdrawn in year 10.

MATH-4:

If you put $100 into a Savings Certificate today paying an Interest rate of 10% per year. How much money will you have in the account after 6 years if no withdrawals are made from it until then, How much of this is Interest?

MATH-5:

If you wish to have $1000 in 10 years from now, how much money must you invest today in a Savings Certificate that pays 8% interest per year?

MATH-6:

If you wish to have $1000 in a Savings Account after 10 years from now an Interest rate of 10%, how much money will you invest in your account each year starting 1 year from now?

MATH-8:

The DELCO Company is attempting to determine the most it should pay to purchase a particular Annuity. The firm requires a minimum return of 8% on all investments. The Annuity consists of cash flows of $700 per year for 5 years.

Time Value of Money Quiz

1. Find the future value at the end of 3 years of $900 invested today at an interest rate of 17 per cent.

a. $1412.85
b. $1426.45
c. $1435.92
d. $1441.45

2. Find the present value of $700 to be received 5 years from today if the interest rate is 8 per cent.

a. $464.02
b. $468.81
c. $476.41
d. $479.45

3. Find the present value of the following cash flow stream if the interest rate is 3 per cent.

0	1	2	3	4	5
	$200	$0	$200	$400	$100

a. $795.4
b. $802.99
c. $815.82
d. $818.86

4. Find the present value of an annuity of $700 per year for 4 years if the interest rate is 10 per cent.

a. $2217.21
b. $2218.91
c. $2220.19
d. $2236.1

5. Find the present value of the following cash flow stream if the interest rate is 3 per cent.

0	1	2	3	4	5	6
	$400	$400	$100	$100	$100	$100

a. $1101.83
b. $1109.83
c. $1115.76
d. $1122.61

6. Find the present value of the following cash flow stream if the interest rate is 12 per cent.

0	1	2	3	4
	$0	$200	$20 m	$200

a. $419.03
b. $428.9
c. $436.26
d. $441.33

7. Find the present value of the following cash flow stream if the interest rate is 15 per cent.

0	1	2	3	4	5	6
	$100	$300	$300	$300	$300	$300

a. $943.13
b. $954.38
c. $961.43
d. $969.5

8. Find the present value of $700 to be recieved 6 years from today if the interest rate is 11 per cent.

a. $361.6
b. $366.25
c. $374.25
d. $382.23

9. Find the future value at the end of 9 years of $1100 invested today at an interest rate of 16 per cent.

a. $4167.84
b. $4181.75
c. $4183.26
d. $4190.16

10. Find the present value of an annuity of $800 per year for 3 years if the interest rate is 15 per cent.

a. $1818.86
b. $1826.58
c. $1830.73
d. $1839.34